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BLOCKCHAIN FOR GOOD ALLIANCE

EMURGO LABS

SDG BLOCKCHAIN ACCELERATOR

Debugging and Testing Report – Green Giraffe

Green Giraffe Agricultural Traceability Platform

Blockchain Integration Report

1. Project Information

Project Name: Green Giraffe Digital Traceability & Market Access Pilot (Kurima Programme)
Challenge & UNDP Office: UNDP Angola
Organization: Green Giraffe Zambia Limited
Document Version: v1.0
Date: 02 February 2026
Network: Cardano Preprod Testnet

2. Overview

The Green Giraffe Digital Traceability System is a comprehensive Laravel-based agricultural supply chain platform integrating Cardano blockchain technology using Aiken smart contracts to deliver immutable traceability, ESG compliance reporting, and carbon footprint verification.

Purpose

This project implements a decentralised agricultural traceability and carbon credit verification system on Cardano. The platform enables:

- Immutable batch registration from farm to retail
- Supply chain traceability across verified stages
- Blockchain-verified carbon footprint calculations and ESG reporting
- Smart Batch Finder with global compliance (EUDR, GlobalG.A.P., FSSC 22000, etc.)
- Life Cycle Assessment (LCA) using Agribalyse 3.2 benchmarks
- NFT-based product verification for consumer trust

Problems Addressed

1. Fragmented and untrusted agricultural supply chains
2. Unverifiable sustainability and carbon claims
3. Difficulty proving compliance with EUDR and international standards
4. Centralised systems vulnerable to tampering
5. Lack of consumer-level verification

SDG Alignment

- SDG 2: Zero Hunger
- SDG 12: Responsible Consumption and Production
- SDG 13: Climate Action

3. System Architecture

```
USER INTERFACES
Farmer | Aggregator | Processor | Transporter | Consumer (QR)

            ↓
Laravel Backend (API + Web)
- Sanctum Authentication
- Role-based Access Control
- Business Logic

            ↓
--------------------------------------------------
```

```
| MySQL | Cardano Attestation | External APIs |
-----------------------------------------------------

            ↓
Blockfrost API (Preprod)

            ↓
Cardano Blockchain (UTxO Model)
- Aiken Validators
  - batch_registry.ak
  - traceability.ak

            ↓
Public Verification
(Cardano Explorer)
```

Data Flow Summary

1. User creates batch/event
2. Laravel validates and stores data
3. Attestation Service builds transaction
4. Transaction submitted via Blockfrost
5. Validators execute on-chain
6. UTxO created with datum
7. tx_hash returned and stored
8. QR scan → public verification

4. Blockchain Design

4.1 Validator: `batch_registry.ak`

Purpose: Immutable batch lifecycle management

Datum Structure

```
type BatchDatum {
  batch_id: Int,
```

```
  batch_number: ByteArray,
  farmer_address: Hash<Blake2b_224, VerificationKey>,
  crop_name: ByteArray,
  crop_variety: ByteArray,
  planting_date: Int,
  expected_harvest_date: Int,
  area_hectares: Int,
  certification_type: ByteArray,
  irrigation_type: ByteArray,
  status: ByteArray,
  created_at: Int,
  data_hash: ByteArray,
}
```

Redeemer

```
type BatchAction {
  CreateBatch { batch_data: BatchDatum }
  UpdateBatch { batch_id: Int, event: BatchEvent }
  VerifyBatch { batch_id: Int }
}
```

Validation Rules

- **Farmer signature required**
- **Minimum 2 ADA locked**
- **Valid timestamps and area**
- **Data hash mandatory**

4.2 Validator: `traceability.ak`

Purpose: Full supply chain traceability

Key Features

- Ordered stage validation
- Quantity continuity enforcement
- ESG and environmental data tracking
- Product recall capability

Minimum Requirements

- ≥3 ADA locked
- Stage sequence enforcement
- Operator signature validation

5. UTxO Transaction Lifecycle

Batch Creation

```
Wallet UTxO → Script UTxO (BatchDatum) → Change back to wallet
```

Traceability Update

```
Previous Script UTxO → Updated Script UTxO
(Value preserved)
```

6. Off-chain Components

Backend

- Laravel 11 / PHP 8.2
- Sanctum + Spatie permissions
- 221 application files

Attestation Service

- Node.js 18 + Lucid
- Port: 8787
- Builds and submits Cardano transactions

Oracles

- NASA HLS (crop health)
- Farm Intelligence APIs
- Agribalyse 3.2 LCA
- Blockfrost API

7. Testing Status

Current: Validators implemented, testing pending

Planned

- Aiken unit tests
- Emulator integration tests
- Preprod testnet validation

Optimisation Targets

- <350k CPU
- <1MB memory
- <8KB script size

8. Tools & Environment

Blockchain

- Aiken CLI 1.9.0
- Lucid
- Blockfrost

Backend

- Laravel 11
- MySQL 8
- PHP 8.2

Frontend

- Tailwind CSS
- Alpine.js
- Chart.js

9. Roadmap

Immediate

- Fund wallets
- Deploy validators
- End-to-end test

Mid-term

- NFT minting
- Reference scripts
- Multi-sig support

Long-term

- Mainnet deployment
- Carbon credit tokenisation
- DAO governance

10. Conclusion

Status: 55% Complete – Ready for Testnet Validation

The Green Giraffe platform delivers blockchain-verified agricultural traceability, ESG compliance, and consumer-level transparency using Cardano and Aiken smart contracts.

Built by Green Giraffe Zambia Limited

Version: 1.0.0
Date: 12 January 2026

Website: https://greengiraffetechnologies.com
Explorer: https://preprod.cardanoscan.io

Appendix A – Quick Reference

Wallets

- **Eternl:** `addr_test1qq96vaw3q9xgcaz7dmkyuphcghx2sva9jn70nrnln63gpv9uem8vlvhnurdy7zpej324gf7zs3kf5mgn48fhsrftzy8s6l64jh`
- **Microservice:** `addr_test1qq9paspj9wr5xz6ru7z505tj8l7zl8x0uw852s58cvclvtew77sr5jj8azt6lmkgx42tzv4w2d9lxgx0737xgysgkxvqn3akhm`

Key Commands

```
npm start
php artisan serve
aiken check
aiken build
```

```
php artisan migrate
php artisan db:seed
```